2.2.5 Synthetic Dyes

Dyes that are produced chemically are called as synthetic dyes. These are classified based on the chemical composition of the dye.

2.2.5.1 Direct Dyes

When a dye colours the fabric directly without the help of any fixing agent, the dye is said to be a direct dye. Direct dyes are water soluble. They are easy to produce, simple to apply and cheap in cost of production and application. Direct dyes are anionic in nature and have greater affinity for cellulosic fibres. They are used to dye cellulose fibres without a mordant in bright shades and they produce a wide range of colours. A levelling agent such as sodium carbonate is added for even dyeing. At the end of dyeing, exhaustion agent such as salt (NaCl) is added which helps the dye to leave the liquor and get attached to the fibre. Some direct dyes are used to dye wool, silk and nylon. Direct dyes can be applied to wide variety of textile materials such as apparel, upholstery fabrics, draperies, linings and automotive fabrics. Most direct dyes have good fastness to light but poor wash fastness.

2.2.5.2 Reactive Dyes

Dyes that react with the fibres and form covalent bonds are called as reactive dyes. They become an integral part of the fibre. They are water soluble and are used to dye cellulose, protein and polyamide fibres. They produce full range of bright shade across the spectrum. They exhibit excellent wash fastness and good light fastness properties. Dyeing of fibre with reactive dyes involves 3 steps, namely exhaustion of dye (NaCl or Glauber's salt), fixation of dye (sodium carbonate or sodium hydroxide) and washing off.

2.2.5.3 Basic Dyes

Basic dyes have cationic or basic groups (positively charged) and hence they are also known as cationic dyes. Basic dyes react with the acidic groups present in the fibres and form electrovalent bonds. Basic dyes are soluble in alcohol but not easily soluble in water. Basic dyes exhibit brilliant shades of colour which is not shown by other dye classes. Basic dyes are suitable for dyeing wool, silk and acrylic, but they have no affinity towards cellulosic fabrics. Basic dyes are used along with a mordant for fibres such as cotton, linen, acetate, nylon and polyester. Basic dyes show moderate light and wash fastness. For dye preparation, the dyestuff is mixed with equal amount of acetic acid followed by warm water under constant stirring.

2.2.5.4 Acid Dyes

Water soluble dyes that require acid (sulphuric, acetic, formic acid etc.,) in dye bath to dye silk or wool are called as acid dyes. These acid dyes are mostly sodium salts of organic acids. When dissolved in water, acid dyes produce negative ions (anions or acidic groups) which react with positive ions of protein fibres and get attached to the fibre through electrovalent bonds. Acid dyes are similar to direct dyes however they cannot be applied to cellulosic fibre due to slight variations in structure. Acid dyes have greater affinity for protein and polyamide fibres. They posses very good fastness to washing and good fastness to light. A large colour range is available with acid dyes. They are inexpensive.

XII TDD_English version CHAPTER 02.indd 24 1/13/2020 2:44:00 PM